Course Outline
FAST School of Management
National University of Computer & Emerging Sciences Lahore Campus

Course:	**Data Analysis for Business I**		
Course code:	MT118	Year/Semester:	1st / 2nd
Program:	BBA/BSAF/BSBA/BFT	Units/Cr Hrs:	3
Department:	FAST School of Management	Instructor:	Aroosa Safdar
Course Type:	Core	Email:	Aroosa.safdar@nu.edu.pk
Pre-Requisite(s):	None	Phone:	042-111-128-128 (325)
Prepared By:		Consultation Hours:	Will be shared in class

COURSE DESCRIPTION

Most decisions in the business world are based both on what is learned in theory and on consideration of real-world observations and experience (i.e. data). Decision-makers at every level of business increasingly are being required to perform their own empirical analysis as personal computers become more advanced, data collection becomes more prevalent, and business decisions become more complex. Thus, a firm grasp of statistical concepts will be crucial to your career. This course introduces you to elementary statistical procedures and reasoning. Whether you will be crunching numbers or interpreting data, this course will make the statistical foundations needed to employ basic methods of sound empirical analysis.

COURSE OBJECTIVES

1	To collect, summarize and descriptively analyze the qualitative & quantitative data.
2	To apply the concepts of probability and probability distributions in real.
3	Perform regression analysis in business context. Able to check the goodness of fit and strength of relationship between two variables in business scenarios.
4	To calculate the indices for prices and quantities of business products.

COURSE LEARNING OUTCOMES

LO#	Learning Outcome Statement	Program Objective	Bloom's Taxonomy Level*
LO1	Discuss data types. Differentiate between population and sample for business data sets. Explain Measurement scales for collection of data. Provide summaries using frequency distributions and graphical representation.	10	Remember, Understand
LO2	Compute and interpret various measures of location and measures of variability. Able to detect outliers.	1,3,5	Understand, Apply
LO3	Use basic counting techniques, laws of probability, specified discrete and continuous probability distributions in various business problems.	1,3,5 & 10	Understand, Apply
LO4	Determine the relationship, fitting a simple linear regression model, and assess the goodness of fit of regression model.	1,3	Understand, Apply
LO5	Calculate and interpret simple price index, aggregate price indexes and weighted price index.	1, 3	Understand, Apply

** Only the highest Bloom's Taxonomy level is mentioned. It is assumed that in order to achieve the higher level, the lower level has been achieved.*

COURSE CONTENTS

LO / Session	Contents
LO1 Session 1-2	• Introduction & Course Outline Discussions • Introduction to statistics, application areas, daily use of statistics, general use in business, types of data. • Types of statistics, population, sample, scales of measurement
LO1 Session 3-6	• Summarizing qualitative data, graphical representation • Summarizing qualitative data, tabular summary • Quantitative data, graphical presentation • Tabular summary of Quantitative data e.g. frequency distributions, stem & leaf plot, cross tabulation
LO2 Session 7 - 12	• Introduction to measures of location & measures of variability • Distribution shapes • Measures of distribution shape & relative location • Detection of outliers using z-scores and Box plot • Measures of Association, weighted mean, grouped data
MID TERM – I	
LO3 Session 13-15	• Introduction to probability, experiments, counting principles, Assigning probabilities & events • Events and their probabilities, some basic relationships of probability • Conditional Probability, Independent events, Multiplication laws
LO3 Session 16-20	• Introduction to Random variables (Discrete & Continuous), The probability distributions for a discrete random variable, properties of expectations, expected value and variance • Specified Probability distributions • Discrete Uniform probability distribution, Binomial Distribution • Poisson distribution • Hyper geometric distribution
MID TERM – II	
LO4-5 Session 21-23	• Continuous Uniform distribution • Introduction to Normal Distribution and its properties • The normal distribution • Normal approximation to the binomial distribution, The exponential distribution
LO6 Session 24-27	• Introduction to Regression analysis, difference between simple linear • regression and multiple linear regression models, determine the simple linear regression equation • Residuals, total variation and its composition, coefficient of determination, correlation coefficient • All the computations in simple linear regression
Session 28	**Review Sessions & Exam Discussion**

** Reading assignments may be also scheduled in the sessions. Please be well-prepared.*

*** Sessions may also involve activities to enhance understanding*

****Course content may change during semester-topics may be added or deleted from the syllabus.*

TEXTBOOK& REFERENCE MATERIAL

Textbook:

- David R. Anderson, Dennis J. Sweeney, Thomas A. Willims.2003 (11th Ed). Statistics for Business and Economics. Ohio, South-Western.

Reference Books:

- Statistical Techniques in Business & Economics 15th ed. by Lind, D.A., Marchal, W. G. Wathen S. A.
- Business Statistics by David M. Levine, Timothoy C. Krehbiel, Mark L.Berenson, P.K. Viswanathan
- Introduction to business statistics by Ronald M. Weiers

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ADMINISTRATIVE INSTRUCTIONS

GRADING POLICY

Grading Mechanism	Weight
Assignments/Project/Quizzes	20%
Class Participation	5%
Mid Term I	10%
Mid Term II	15%
Final Exam	50%

Grading Criteria			
	ABSOLUTE Grading	☒	**RELATIVE Grading**
***** Final Grading will be relative grading, using inter-quartile range/MCA method.			

Class Policies

- Please note that any exception for one student is unfair to all other students, so don't expect any.
- Please turn off and store away cell phones, Ipads, laptops, and other electronic devices.
- Talking during lecture is not permitted. It is disrespectful and disruptive to other class members and the instructor.
- If you miss a class, it is your responsibility to determine what was covered including any administrative announcements.

Studying

The proper way of studying for this class is following ADA policy; a short description of the same is as follows

- Ahead of the class, it is expected that you have read the relevant chapters from the textbook;
- During the class you are expected to follow the lecture, take notes and ask questions; and
- After every class you would review your notes and solve the end of chapter exercises and read the textbook.

An extensive set of practice problems is placed on the Xeon Server, make sure you cover them all. We might set a tutorial session from time to time to answer your queries about the problems/practice sets.

Attendance Policy

As you can see, almost every other session is a different topic and as such missing any classes will result in huge setback as you will lose substantial course content. Besides, owing to the interactive nature of the course, attendance is crucial. Therefore, only under unavoidable circumstances, student should think of missing a class. *See university policy on minimum attendance allowed for final exams.*

Scholastic Dishonesty

In the classroom and in all other academic activities, students are expected to uphold the highest standards of academic integrity. Any form of scholastic dishonesty is an affront to the pursuit of knowledge and jeopardizes the quality of the degree awarded to all graduates.

What exactly is plagiarism?

- Plagiarism is a form of cheating.
- Plagiarism is using someone else's ideas or words and saying they are your own.

If you use material from a text and do not acknowledge the source, you are committing plagiarism.

Specifically, these behaviors are often regarded as plagiarism:

- Copying directly from a text, acknowledging the source but pretending that you are paraphrasing.
- Paraphrasing or copying directly from a text without acknowledging the source.
- Copying from another student's assignment with or without the student's knowledge.

The following behaviors are regarded as misconduct:

- Submitting the same assignment in two different papers.
- Getting someone else to write an assignment for you.

You are also involved in misconduct if you:

- Let another student copy from your own work.
- Write an assignment for another student.

Students who violate the rules on scholastic dishonesty are subject to disciplinary penalties, including the possibility of failure in the course and/or dismissal from the University. *Since dishonesty harms the individual, all students, and the integrity of the University, policies on scholastic dishonesty will be strictly and actively enforced.*